DATA ANALYSIS PROJECT

VIDEO GAME SALES

EXPLORATORY Data Analysis

1. Introduction:

In this exploratory data analysis project, we examine a dataset containing video game sales across different regions and platforms. Our work begins by exploring the dataset, understanding its structure, and identifying issues such as missing values and duplicate records. By cleaning and organizing the data, we aim to uncover meaningful patterns in global sales across different game genres and platforms. We also examine the relationship between critic scores and sales figures to better understand the patterns within the dataset.

2. Data Cleaning:

In this project, we inspected the dataset to understand its structure, identify missing values, and check for duplicate records. We used the Python library pandas, which provides useful functions for data cleaning and manipulation. We began by inspecting the dataset using head(), reviewing its information with info(), and checking the data types of each column using dtypes.

We removed duplicate rows using the drop_duplicates() function. Furthermore, we identified missing values, specifically in the Critic_Score column, where approximately 25.68% of the values were missing. We addressed this issue by replacing the missing values with the median.

We also recalculated Global_Sales by summing the sales across the four regions: Japan, North America, Europe, and the rest of the world. The resulting values were rounded to two decimal places to maintain consistency.

Finally, we saved the cleaned dataset to a new file named vgsales_cleaned.csv.

3. Exploratory Data Analysis:

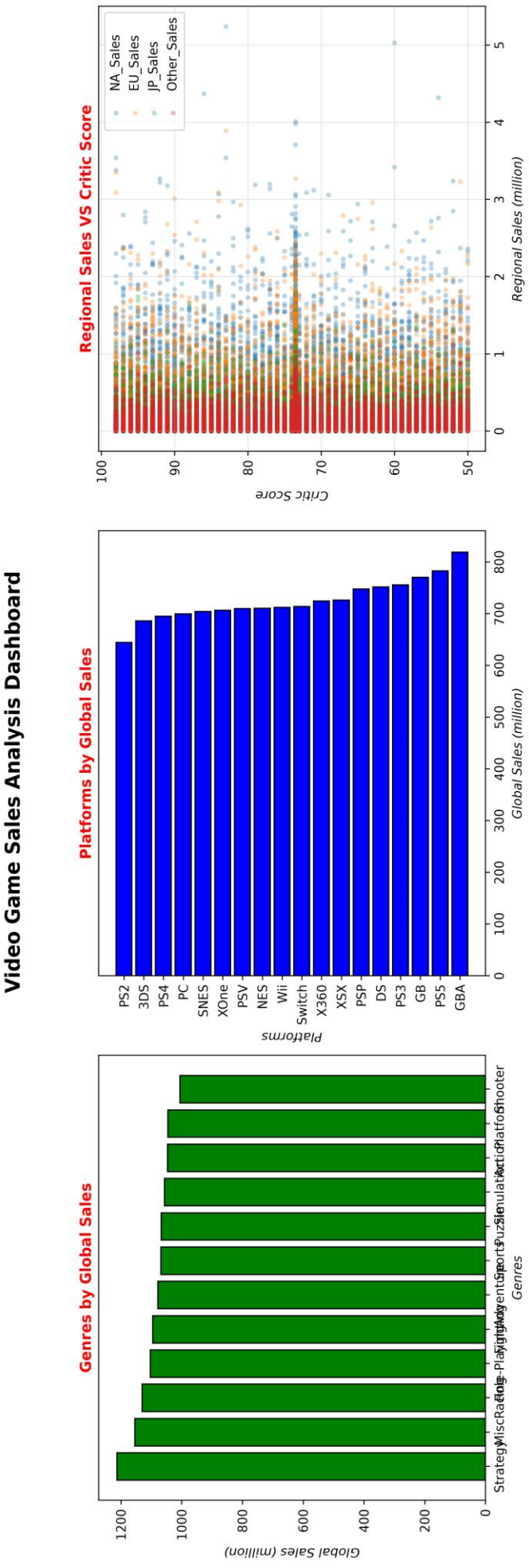
Once we had prepared the dataset, we began exploring it to identify meaningful patterns in video game sales. We started by identifying the best-selling game genres globally. Strategy games ranked first, with approximately 1.21 billion in global sales. Shooter games recorded approximately 1.01 billion, while other genres, including Misc, Racing, Role-Playing, and Fighting, also achieved substantial sales figures.

In the next phase, we examined the linear correlation between sales across different regions—Japan, North America, Europe, and the rest of the world—and critic scores. We also examined the relationship between critic scores and global sales. The results showed a very weak linear correlation, with the global sales correlation coefficient being approximately 0.0151. This suggests that no clear linear relationship was observed in this dataset. It is worth noting that missing critic scores had been replaced with the median during data cleaning.

Next, we identified the best-selling gaming platforms based on their total global sales. The GBA (Game Boy Advance) ranked first, with 819.35 million in sales. It was followed by the PS5, with 783.10 million, while the PlayStation 2 recorded the lowest total among the platforms in our dataset, at 644.32 million.

Finally, we implemented a simple interactive search feature. By entering a game title or part of its name, users can retrieve matching records, view the number of matching records, and calculate their combined global sales. This feature provides a convenient way to explore the sales information available for specific games in the dataset.

4. Data Visualization:



5. Insights & Findings:

Through this project, we identified several important patterns in the dataset. Our analysis showed that total global sales vary across game genres and platforms, with Strategy ranking first among genres and GBA ranking first among platforms.

These differences highlight how sales performance varies across different categories and platforms. However, our analysis focuses on identifying patterns within the available data rather than determining the exact factors responsible for these differences.

Furthermore, we observed a very weak linear correlation between critic scores and global sales. This indicates that no clear linear relationship was identified between these two variables in the dataset. However, correlation alone is not sufficient to determine whether critic reviews directly influence sales.

6. Conclusion:

In conclusion, this project allowed us to explore video game sales through a structured data analysis process. We moved from data inspection and cleaning to identifying meaningful patterns and presenting our findings through visualizations.

Throughout this project, we used Python libraries such as Pandas for data manipulation and analysis, and Matplotlib for data visualization. We also developed our analytical thinking by formulating questions, examining relationships, and interpreting the results.

In future projects, we aim to further improve our analytical methods, expand our technical skills, and create more informative visualizations. Our goal is to produce reliable, data-driven insights that can support better decision-making.

Thank you for reading this document.

Peace, mercy, and blessings of God be upon you.